

Section A Data Fundamentals

1. What is database?

A database is an organised collection of information stored electronically so that it can be easily accessed, managed, updated and analysed. They help organisations and business store large amounts of information without relying on ^{separate} paper records or scattered files.

The databases are essential in the modern world because businesses and organisations generate large amounts of data every day. The databases makes it easier to find information, keep records accurate and use the information to make better decisions.

An example of databases A university using the database to store information about student records, course enrollment test results.

Another example is South African Forestry Company Limited (SAFECOL) using database to store information about the forestry plantation (Compartments, Tree species, planting dates, Tree measurements, harvesting records and forest health). A forester then uses this information to monitor forest resources and use it to plan management activities.

2. Data warehouse - This is a central system used to collect and store large amounts of data from different sources for reporting and analysis.

Data warehouse differs from a regular database by that the regular database is normally designed to handle day to day activities such as adding or updating records. A warehouse is normally designed for analysing ~~historing~~ historical data and identifying trends.

A regular operational database is designed for fast, frequent read and write operations while a warehouse is designed to store large amounts of data, it is usually updated in batches rather than continuously.

A company would likely use data warehouse to combine data from sales, marketing and finance system into one place so people in charge can run reports and make decisions based on full picture.

3. Data Lake - This is a storage that can hold large amounts of data in its original form. The data can include structured, semi-structured and unstructured information. until it is needed.

Data lake is different from a warehouse since a warehouse stores cleaned, structured, organised data while data lake stores raw data as it is and can later then decide how to structure it only when you use it (Schema-on-read).

A business would prefer a data lake when it has to store large volumes of varied data and expects to analyse the data in a different ways later. For examples (storing Excel inventory files, GIS data, satellite images and photographs of plantations, GPS information) for a forestry company.

4. Data mart - This is a smaller, focused collection of data designed for a particular department, business area or group of users. It can be created from a larger data warehouse. Instead of giving every employee access to all the organisation's data, a department can have access to the information relevant to its work. Departmental analyst and managers who only need data relevant to their area. For example: HR department, marketing, finance and operation).

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Structured data - Is data organised, to fit in a fixed format normally in rows and columns. It makes it easy to store and analyse in a database.

Example: Forest Inventory spreadsheet containing columns about data recorded on tree species, DBH, Tree height, and plot number.

Semi-structured data - This type of data does not follow a strict table but contains some organisation through tags, keys, or other markers.

Example: A JSON file containing information about different forest compartments.

Unstructured data - It does not have any predefined table-like structure.

Example: Photos, Video taken during forest surveys such as images of pests and diseases or fire damage.

6. Data modelling

This is a process of designing how data can be organised, stored and related to other data within a system, creating a blueprint before building the database.

It is important because it helps ensure that information is organised logically, reduces unnecessary duplication and makes it easier to retrieve and analyse data.

2 types of data modelling

- * Relational data model: It organises data into tables containing rows and columns. Different tables can be connected using relationships.

- * Hierarchical data model - It organises information in a tree-like structure where records have parent and child relationships.

7. Data mining - This is the process of mining

- * examining large amounts of data to discover useful patterns, relationships and trends that may not be obvious. Businesses use data mining to support decision-making, understand customers, identify problems and predict possible ~~future~~ future outcomes.

Examples * Banks like ABSA and Standard bank using data mining to detect unusual transaction patterns that may indicate fraud.

- * Clicks and Dischem retailers can use data mining to analyse purchase history to discover that customers who buy diapers (Pampers, Huggies) also buy baby wipes, so they place them near each other.

8. Database management system (DBMS) - This is a software that allows users to create, store, organise, retrieve and manage data in databases.

MySQL - It is an open-source relational database systems suitable use; For websites and business application that need structured data.

PostgreSQL - An open-source relational database that is known for its powerful features.

Suitable use in complex business, research and analytical databases.

Oracle Database - A powerful commercial relational database system suitable for large organisations that manages very large and important databases (Banks & Universities).

MongoDB - A NoSQL database that stores information in flexible documents formats, it is suitable for applications that does not always follow a fixed table structure.

9. ETL (Extract, Transform, Load) - ~~What is~~ This is a process used to move data from source systems into data warehouse.

- Extract - Data is collected from different sources, such as database, excel files, CSV files and other system.
- Transform - The data is then cleaned and changed into a suitable format, this may include removing duplicate, correcting errors, standardising names and dealing with missing values.
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Load - The cleaned data is now placed into a target system such as data warehouse where it can be analysed.

The ETL is important because it ensures that the data used for analysis is accurate, consistent and usable.

10. Common data storage formats used in data analytics

CSV (Common-Separated Values)

It stores data in a simple table-like format where values are separated by commas. It is easy to create, read and import into programs such as excel and python.

JSON (Javascript Object Notation) - Stores data in a structured format using key-value pairs. It is commonly used for APIs and transferring data between applications.

Parquet - It is a column-based file format that is designed for efficient storage and analysis of large datasets. It is particularly useful when working with large amounts of data.

Avro - A data serialization format that stores data together with its schema. It is often used when data needs to be transferred between different systems.

Use CSV for small, simple data sharing, JSON for web/API data, Parquet for large scale analytics, Avro for streaming/pipeline systems needing schema evolution.

Section B Artificial Intelligence and Machine Learning

11. Artificial Intelligence - Is the technology that allows computers and machines to perform tasks that normally require human intelligence. These tasks can include learning, recognising patterns, solving problems, understanding language and making decisions.

Narrow AI (Weak AI) - This is AI that is designed to perform a specific task or a limited range of tasks.
For example: A spam filter and Siri

General AI (Strong AI) - It refers to a theoretical form of AI that would be able to understand and perform a wide range of intellectual tasks at a human-like level. It currently does not exist.
Example of such AI if it was to exist would be an AI that can learn, reason across all domains as a human.

12. Machine learning - This is a part of AI in which computers learn patterns from data and use those patterns to make predictions or decisions without having every rule specifically programmed by a person.

The difference between traditional and machine learning is that in TL a programmer provides the rules and data to produce an output while in machine learning the computer is given data and examples and it then learns patterns that can be used to produce predictions or decisions.

Main types of ML

Supervised - This type of ML model learns from labelled data where the results are already known.
examples - Predicting the prices of poles/wood to be sold.

Unsupervised - This model finds patterns in unlabelled data and tries to discover patterns or groups within the data.

example - Grouping customers into segments based on shopping behaviour at Shoprite.

Reinforcement learning - This model learns from trial and error, receiving rewards or penalties.
example - AI learning to play chess by playing many games and being rewarded.

13. Deep learning - This is a specialised part of ML that uses neural networks with multiple layers to learn complex patterns from large amounts of data like images, audio and texts.

The relationship between AI and ML is that AI is the broader field while machine learning is one approach within AI, while deep learning is a more specialised approach within machine learning.

Neural Networks are computing systems that are made up of connected nodes, sometimes called artificial neurons. They generally contain an input layer, one or more hidden layers and an output layer.

Information passes through these layers and the models adjust its internal parameters during training so that it becomes better at producing the desired output.

14 Natural language processing - This is a branch of AI that enables computers to understand, process and generate human language including written text and speech.

Examples of NLP

* Voice Assistants (Siri and Alexa), we use them to understand spoken commands and questions from our phones.

* Chatbots - These are customer-service bots that can understand questions written by users and provide responses.

* Search engines - They use language processing techniques to understand what a person is asking and provide relevant results.

~~15~~ Language

15 Large language model (LLM) - An AI model that is trained on very large amounts of text so that it can recognise patterns in language and generate or work with human-like text.

The LLM differs from a standard machine-learning model because it is generally designed for a broad range of language tasks rather than one narrow prediction task. It can answer questions, summarise information and assist with coding.

examples

1. ChatGPT - This is an AI system that can interact with users using natural language and assist with tasks such as explaining concepts, writing, summarising and coding.

2. Claude - An LLM based AI assistant that can work with natural language instructions and assists with tasks such as analysis, writing and coding.

16. Generative AI - A type of AI that can create new content based on instructions and prompts. This content can include text, images, audio and video.

At high level the generative AI learns patterns from large amounts of training data. Normally when I provide the prompt the AI then uses the instruction to generate the output.

examples of generative AI

* ~~Am~~ Lovable - A generative AI powered software that helps generates codes

* ChatGPT - It assist with generating and working with text, codes and analysis.

* AI image generator - It is used to generate images from prompts

17. Training database - A training database is a collection of data used to teach a machine-learning or AI model how to recognise patterns and perform a particular task. The quality of training data is important because an AI model learns from the information provided to it. If the data is inaccurate, incomplete or biased

the model can learn incorrect patterns and produce unreliable results.

If the training data is biased or incorrect, the model may produce biased predictions, inaccurate classification or unfair decisions.

16. Computer vision - This is a field of AI that allows computer to process and interpret visual information such as images and videos. Then machine learning can be used to help computers identify objects, patterns and other information, they do this by breaking down images into pixels, algorithms detects edges, shapes and patterns to identify objects.

Examples :

- * Facial recognition on our Smartphones to unlock.
- * Hospital scanners like MRI machines that analyse medical images to help identify signs of diseases.

Section C APIs, Tools and Developer Concepts

19 Application Programming Interface (API) - It is a set of rules that allows different softwares applications or systems to communicate with each other and exchange information.

For example in forestry there is what we call Fire danger Index as part of Forestry Fire management system, there is a system we use to calculate the FDI, needing information such as temperature, wind direction and humidity. Instead of collecting this information itself, an API can be used to request weather data and then ~~used the~~ API sends information to FMS which then calculate the Fire danger Index.

APIs are important because they allow different systems to share information and work together without having to build all the functions themselves.

20 API keys are a unique code used to identify and authenticate an application or user when accessing an API. Its purpose is to track usage, control access and can enforce rates limits ~~and~~ or billing. monitoring how much it is being used.

If the API key gets exposed publicly, another person may use it to make requests. increasing costs on your account, and accessing private information.

The API must be stored and kept private. The key codes should not be posted publicly on websites and Github.

21 REST API - This is an API that follows a set of principles for allowing applications to communicate over the web.

The acronym REST stands for Representational State Transfer.

The main HTTP methods used to perform different operations are

HTTP method	Their purpose	Examples
GET	To retrieve information	Use to get information about forest compartments
POST	It creates or submit new information	Adding new Forest inventory method.
PUT	To update or replace existing information	To update and remove old compartment on compartment list.
DELETE	Removes my	
DELETE	Removes information	Deleting duplicate information entered on the comp list

22 JavaScript Object Notation (JSON) - This is a lightweight text format used for storing and exchanging data using key-value pairs.

The JSON is commonly used with APIs because it is relatively easy for both humans and computers to read and process.

example of JSON :

```
{  
  "name": "Zwivhuya",  
  "Age": 28,  
  "CurrentProfession": "Forester",  
  "Career Goal": "Data analyst"  
}
```

23 Prompt engineering - This is the process of designing clear and effective instructions or prompts, for an AI system so that it produces a more useful response. Prompt skill is important because the quality and clarity of the instructions can affect the usefulness of the output since Chat and Claude responds based on how a request is phrased.

Example of Poor and Improved prompt

Poor prompt	Improved prompt
Analyse my data	Please help me analyse this forest inventory dataset and identify the most common tree species. Explaining the results in simple terms and present them on a table.
Write python code on DBH	Write beginner friendly python code using pandas to calculate the average DBH (Diameter at Breast Height) for each tree species in my forest inventory dataset.

24 Map Function - This is a programming function that applies the same operation to each item in a collection such as a list or array. After that it then produces a new collection containing the results.

example of map function: using python for tree height conversion from metres to centimetres.

```
heights = [10, 12, 15]
heights_cm = list(map(lambda x: x * 100, heights))
print(heights_cm)
```

25 Cloud computing - This means using the computing resources such as storage, servers, database and software through the internet instead of relying entirely on a computer local hardware.

IaaS - Infrastructure as a Service - This cloud service provides basic computing infrastructure such as virtual servers, storage and networking.
Example: Amazon EC2

PaaS - Platform as a service - It provides a platform that developers can use to build and deploy applications without managing all the underlying infrastructure.
example: Google App Engine

SaaS - Software as a service - The SaaS provides complete software application through the internet.
example: Google workspace.

26 Version control system - A VCS is a system that records changes made to file over time. It allows users to track previous versions, identifying changes and return to an earlier version when necessary.

Git is a popular version control system that is widely used by developers and data analyst because it allows them to track changes to code, analysis files and projects. Using GitHub as a platform to host GIT repositories.

The Git concepts are as follows.

Commit: A commit is a saved record of changes made to a project. It

Branch: A branch is a separate line of development within a project. It allows someone to work on a new feature or change without immediately changing the main version of the project.

27. Jupyter Notebook - This is an interactive web-based tool for writing and running codes in small sections alongside text, visualisation and notes.

It combines programming code, explanation, data and visualisation in one document. It is most commonly associated with Python but it does support other programming languages.

Jupyter Notebook is popular among data scientists and analysts because they can write and run codes step by step, analyse datasets, create charts and visualisation, add explanation using text, documents the process used to reach a result.

28. Python - This is a high-level general purpose programming language that is widely used in areas such as data analytics, AI, machine learning, automation and software development. Python is the most popular language for data analytics because it is relatively easy for beginners to understand and it has got many libraries that are specifically designed for working with data.

The 4 Python libraries commonly used in data work are

Pandas - Used working with, cleaning and analysing structured datasets such as CSV files.

Numpy - Used for numerical calculations and working with arrays and mathematical data

Matplotlib - Used for creating graphs and data visualisation

Scikit-learn - Used for building and evaluating machine-learning models

Section D: Data Ethics, Governance and Careers

29. Data privacy - This refers to the proper handling and protection of people's personal information. It involves ensuring that personal data is collected, stored, used and shared responsibly and only for appropriate purposes.

It is important because personal information can be misused if it is not properly protected. People must always have an understanding of how their information is being collected and used.

The General Data Protection Regulation (GDPR) - Is a data protection law of the European Union. It establishes rules for how organisations handle the personal data of individuals.

The GDPR law applies to organisations operating in the European Economic Area and can apply to organisations outside the EU/EEA when they offer goods or services to people in the EU.

examples: When collecting data on household use of forest resources the personal information must always be protected and used only for specific research purposes.

30. Data Ethics refers to the moral principles and responsible practices involved in collecting, analysing, storing and using data. It is important because it guides how data must be collected so that animals, environment, systems are protected.

Examples of unethical use of data

1. Using personal information without permission

It is problematic because the information stolen can be used for purposes that they did not agree to and it violates people's privacy.

2.

2. Using biased AI to make decisions

Problematic because organisations can use this information on AI systems to make decisions about people when the system has been trained on biased data.

3. Selling user data without consent

It is problematic because it violates people's trusts and privacy rights.

31. Data governance

Data governance is a system of rules, responsibilities, processes and standards used by an organisation to manage its data properly. The main purpose is to make sure that data is accurate, secure, consistent, accessible to authorised users and used appropriately.

The key components of a good data governance framework is includes

- * Data quality: Ensuring that data is accurate, complete and reliable
- * Data security: Protecting data against unauthorised access/loss
- * Data privacy: Protecting personal and sensitive information.
- * Roles and Responsibilities: Identify clearly who is responsible for managing different types of data

Data standard: Establishing consistent ways of collecting, recording and storing data.

Data access: Ensuring that people can access the information they need while preventing unauthorised access

32 Bias in AI - Normally occurs when AI system produces systematically unfair or inaccurate results because of problems or imbalance in its training data, design or use. A real world example is when some of the facial recognition systems have historically performed less on darker skin tones because they were trained mostly on lighter skin tone

The consequences of biased AI includes misidentification, unfair treatment and reinforcement of existing social inequalities.

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Career	Roles	Skills	Common Tools
Data analyst	Examine data to identify trends and provide information that support decision	SQL, Excel, Statistics, Data Visualisation	Excel, SQL, Power BI, Tableau, Python
Data Scientist	Uses statistics, programming and machine learning to analyse data and build predictive models	Python, Statistics, Machine learning, mathematics	Python, Jupyter, Sci, KIT-learn, SQL

Career	Main Role	Skills	Common Tools
Data Engineer	Builds and maintain systems that collect, transforms and store data	Programming, database, ETL, Cloud technologies	SQL, Python, Spark, cloud Platform
AI Engineer	Develop and implements AI and machine learning systems in applications	Machine learning, Programming, AI, Model development	Python, TensorFlow/PyTorch, APIs, Cloud platform

B4. Business Intelligence - Refers to the use of data, analytical tools and technologies to help organisations understand their performance and make better decisions. The BI systems take data from different sources and turn it into reports, dashboards, charts and other visualisation that make information easier to understand.

The most popular BI tools are

* Power BI - It is a business - Intelligence and data visualisation tool that allows users to connect to data analyse it and create interactive reports and dashboards

→ Use Power BI to create dashboard that shows plantation area, tree species and harvesting process

* Tableau - Data visualisation and BI platforms that helps users explore data and create interactive visualisation and dashboards

They help business because visualising data makes it easier to identify trends, compare performance and make evidence based decision.

Section E: Personal Reflection

35 I found the Generative AI most interesting because it can create new content such as text, images, codes and even applications.

36 The APIs, REST and some of the programming concept were the most difficult for me since it was my first time hearing about them. In order to help myself understand more I started studying them with practical examples relating them to everyday use and using online tools.

37 As someone who is transitioning or having interest in AI and data analysis, I see this transitioning helping me merge my Forestry degree and Data into a career I can venture into. And also will be able to use tools such as Python, SQL and Power BI to help me turn my data into useful insights and visualisation.

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